

SolarMax

Pre-Deployment Checklist — Point It at the Sun

Four numbers to check before the panel goes outside.

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1. Before you start

This assumes you’ve already **flashed** the board and **set the clock** (see the Clock Setup guide). This guide covers the last step: making sure the panel actually points at the *right* place when you power it up outside.

Note

How the tracking works: SolarMax computes where the sun is purely from the **date, time, and location** — there are no light sensors steering it. So on power-up it works out the sun’s position and drives the panel there automatically. That means it can only be right if the **location** and the **mounting angle** are correct (both in `include/config.h`), and the **limit switches** are working.

Note

Position is tracked by the motor and limit switches — no calibration step for you. On power-up the panel homes to the limit switches and measures its own travel speed automatically (you’ll see a `[MOTOR] Calibrated: ... deg/s` line). The potentiometer is **not** used for this anymore — it stays wired but does nothing for control.

2. Check 1 — Location (config.h lines 4–5)

Open `include/config.h` and set your site’s coordinates on **lines 4 and 5**:

```
#define LATITUDE      33.4255f    // Decimal degrees North
#define LONGITUDE     -111.9400f   // Decimal degrees (negative = West)
```

The defaults are Tempe, AZ. Change them to wherever the panel will actually sit.

Tip

Getting your coordinates: in Google Maps, right-click your exact spot — the latitude and longitude appear at the top of the menu (e.g. 33.4255, -111.9400); click them to copy. **North latitude is positive; West longitude is negative** (keep the minus sign). Keep the trailing `f`.

3. Check 2 — Mounting angle (config.h lines 86–87)

These describe how the rotation *axis* is mounted, on **lines 86 and 87**:

```
#define AXIS_TILT_DEG      0.0f    // Degrees above horizontal (0 = flat roof)
#define AXIS_AZIMUTH_DEG  0.0f    // 0=North, 90=East, 180=South, 270=West
```

- **Standard install (flat and level, axis running North–South):** leave both at `0.0f`. This is the common case — if in doubt, use this.
- **If the mount is tilted** (sloped roof, or a seasonal tilt): set `AXIS_TILT_DEG` to the pitch angle (degrees above horizontal), and `AXIS_AZIMUTH_DEG` to the compass direction the *raised* end of the axis faces.

Note

Getting these wrong tilts the whole day's tracking off. If you're not tilting the mount, `0 / 0` is correct. The full geometry is in `docs/sun_tracking_math.md` if you need it.

4. Check 3 — Limit switches (position calibrates itself)

There is **no manual calibration** here. Position is tracked by the motor and the two limit switches: on power-up the panel homes to the limits and measures its own travel speed automatically. Your only job is to confirm the **limit switches actually work**.

When you power up (Serial Monitor open) you should see, within a minute:

```
[MOTOR] Homing to east limit...
[MOTOR] Sweeping to west limit to measure travel speed...
[MOTOR] Calibrated: 60 deg in 18.4 s = 3.26 deg/s
```

Seeing the **Calibrated** line means both limit switches tripped and the panel knows where it is. You're done — nothing to edit.

Important

If instead you see **never reached east/west limit** --- check limit-switch wiring, a limit switch isn't triggering. The tracker will **not** move until this is fixed — the limit switches are what tell it where the panel is. Check their wiring (and `LIMIT_ACTIVE` in `config.h` if you use normally-closed switches).

Note

The panel does a slow home-and-sweep every time it boots. In the field it never powers off, so you'd only ever notice this while (re)programming.

5. Finish — upload and power up outside

1. Re-flash the real firmware after your edits:

```
pio run -e esp32dev -t upload
```

2. Power the panel up **during daylight**. Within a few seconds it computes the sun's position and drives the panel to face it, then nudges it westward through the day. In the Serial Monitor you'll see lines like:

```
[SUN] Local 10:15 El=42.7 deg Az=131.6 deg Panel target=-8.4 deg
```

Tip

Plugged in at night? That's fine — it parks and prints `[NIGHT] Waiting for sunrise...`, then starts facing the sun automatically at sunrise (sun above `SUN_MIN_ELEV_DEG`, `config.h` line 94). High wind ($\geq$ `WIND_STOW_MPH`, `config.h` line 92) sends it flat to stow.

6. Quick checklist

- ☐ Clock set (Clock Setup guide) — no `No valid time source` error.
- ☐ **Latitude** correct — `config.h` line 4.
- ☐ **Longitude** correct (minus sign if West) — `config.h` line 5.
- ☐ **Axis tilt** matches the mount (0 if flat) — `config.h` line 86.
- ☐ **Axis azimuth** matches the mount (0 if flat) — `config.h` line 87.
- ☐ **Limit switches work** — saw `[MOTOR] Calibrated: ... deg/s` on boot (Check 3).
- ☐ Re-flashed `esp32dev`, powered up in daylight, panel faces the sun.

7. Troubleshooting — “it moves, but to the wrong place”

- **Doesn't move at all, logs never reached...limit or not calibrated:** a limit switch isn't triggering, so the panel never homed — check the limit-switch wiring (Check 3).
- **Panel drifts a few degrees off during the day:** normal for open-loop timing — it re-zeroes itself at the limit switches each morning and evening.
- **Tracking is off by a consistent amount all day:** the mounting angle is wrong — recheck Check 2 (`config.h` lines 86–87), and that the axis really runs North–South.

- **Off by hours / points east when it should point west:** location is wrong — recheck Check 1 (config.h lines 4–5), especially the longitude sign.
- **Doesn't move at all, logs No valid time source:** the clock isn't set — see the Clock Setup guide.
- **Sits flat and won't track:** it's in stow (wind) or night — check the wind reading and that the sun is actually up.